

# MATH 6112. FALL 2026. ASSIGNMENT 1.

**Zejia Chen**  
**GTID: 904178159**

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## Problem 1

### Exercise 1.3.13-1

Suppose there are two different minimal polynomials  $p_1, p_2$  of  $T$ . We use  $\deg f$  to denote the degree of polynomial  $f$ . By minimality,  $\deg p_1 = \deg p_2$ . Since both  $p_1$  and  $p_2$  are monic, we have that  $\deg(p_1 - p_2) < \deg p_1$  and  $(p_1 - p_2)(T) = p_1(T) - p_2(T) = 0$ . Since  $p_1 - p_2 \neq 0$ , it contradicts the minimality of  $p_1$ . Therefore, the minimal polynomial is unique.

### Exercise 1.3.13-2

Let  $p = a_0 + a_1x + \cdots + a_kx^k$  be the minimal polynomial ( $a_k = 1$ ). It is trivial that  $k \geq 1$ .

$\implies$  : Suppose  $a_0 = 0$ . Then we have that

$$0 = T^{-1}p(T) = \sum_{i=1}^k a_i T^{i-1} =: q(T)$$

where  $q(x) = a_1 + a_2x + \cdots + a_kx^{k-1}$ .  $q \neq 0$  since  $k \geq 1$ .  $q$  has lower degree than  $p$  and  $q(T) = 0$ . This contradicts the minimality of  $p$ . Hence,  $a_0 \neq 0$ .

$\Leftarrow$  : Since  $p(T) = 0$ , we have that

$$T \left( - \sum_{i=1}^k \frac{a_i}{a_0} T^{i-1} \right) = I.$$

We complete the proof by setting  $T^{-1} = - \sum_{i=1}^k \frac{a_i}{a_0} T^{i-1}$ .

### Exercise 1.3.13-6

(a):  $Tv = \alpha v$  implies that  $0 = T^k v = \alpha T^{k-1} v = \cdots = \alpha^k v$ . Since  $v \neq 0$ ,  $\alpha^k = 0$  shows that  $\alpha = 0$ .

(b): WLOG, we may assume  $p = k - 1$  ( $T^k = 0$  and if  $p < k - 1$ , we set  $\alpha_i = 0$  for  $p < i \leq k - 1$ ). Construct  $\{\beta_i\}_{i \in [k-1] \cup \{0\}}$  as follows,

$$\beta_0 = \frac{1}{\alpha_0} \quad \text{and} \quad \forall i \in [k-1], \beta_i = -\frac{1}{\alpha_0} \sum_{j=1}^i \alpha_j \beta_{i-j}. \quad (1)$$

Set  $P = \sum_{i=0}^{k-1} \beta_i T^i$ . By Equation (1),  $SP = I + T^k H = I$ , where  $H$  is some polynomial of  $T$ . Therefore,  $S$  is invertible and  $S^{-1} = P$ .

## Problem 2

We first prove that  $C$  is invertible. Let  $V = (v_1, \dots, v_n)$  and  $W = (w_1, \dots, w_n)$ . Let  $D$  be the matrix associated to the linear transformation which changes the basis:  $v_j = \sum_{i=1}^n d_{ij} w_i$ . We have that  $VC = W$  and  $WD = V$ . Therefore,  $VCD = WD = V$ . Since  $v_1 \dots v_n$  are linearly independent,  $CD = I$  and  $C$  is invertible.

We then prove that  $B = DAC$ . The following holds:

$$\begin{aligned}
 WDAC &= VAC = [T(v_1), \dots, T(v_n)]C && (\text{def. of } A) \\
 &= \left[ \sum_{i=1}^n c_{i1} T(v_i), \sum_{i=1}^n c_{i2} T(v_i), \dots, \sum_{i=1}^n c_{in} T(v_i) \right] \\
 &= \left[ T\left(\sum_{i=1}^n c_{i1} v_i\right), T\left(\sum_{i=1}^n c_{i2} v_i\right), \dots, T\left(\sum_{i=1}^n c_{in} v_i\right) \right] && (T \in \text{Hom}(V, V)) \\
 &= [T(w_1), \dots, T(w_n)] = WB. && (\text{def. of } B)
 \end{aligned}$$

Since  $w_1, \dots, w_n$  are linearly independent, we have that  $DAC = B$ .

## Problem 3

(a): By Block Diagonal Form (Corollary 1.5.45), there exists  $V$  such that  $V^{-1}AV = R = \text{diag}\{R_{ii}\}_{i \in [p]}$ , where  $R_{ii} = \lambda_i I + U_{ii}$  and  $U_{ii} \in \mathbb{C}^{m_i \times m_i}$  is strictly upper triangular. Fix  $k \in [p]$ , let  $U_{kk} = (u_{ij})_{i,j \in [m_k]}$  and  $v_1 \dots v_{m_k}$  be the  $(1 + \sum_{i < k} m_i)$ -th to  $(\sum_{i \leq k} m_i)$ -th column vectors of  $V$ . Since  $AV = VR$ , we have that

$$\begin{cases} Av_1 = \lambda_k v_1 \\ Av_2 = u_{12}v_1 + \lambda_k v_2 \\ \vdots \\ Av_{m_k} = \lambda_k v_{m_k} + \sum_{i < m_k} u_{im_k} v_i \end{cases} \implies \begin{cases} (A - \lambda_k I)v_1 = 0 \\ (A - \lambda_k I)^2 v_2 = u_{12}(A - \lambda_k I)v_1 = 0 \\ \vdots \\ (A - \lambda_k I)^{m_k} v_{m_k} = (A - \lambda_k I)^{m_k-1} \sum_{i < m_k} u_{im_k} v_i = 0 \end{cases}$$

Therefore,  $v_1, \dots, v_{m_k}$  are solutions of  $(A - \lambda_k I)^{m_k} v = 0$  and the solution set contains at least  $m_k$  linearly independent vectors. On the other hand, let  $v$  be one column vector of  $V$  and  $v \notin \{v_1, \dots, v_{m_k}\}$ . Suppose  $\lambda$  be the corresponding eigenvalue of  $v$ , we have that

$$(A - \lambda_k I)^{m_k} V = V(R - \lambda_k I)^{m_k} \implies$$

$$(A - \lambda_k I)^{m_k} v = (\lambda - \lambda_k)^{m_k} v + \text{linear combination of column vectors of } V \text{ except } v \neq 0,$$

since  $\lambda - \lambda_k \neq 0$  and column vectors of  $V$  are linearly independent. Since  $\text{rank}(V) = n$ , the dimension of solution set of  $(A - \lambda_k I)^{m_k} v = 0$  is  $m_k$ .

(b) Putting together the sets of vectors from (a), we get exactly  $V$ . Since  $\text{rank}(V) = n$ , it is a basis of  $\mathbb{C}^n$ .

## Problem 4

(a) Suppose  $v_1, \dots, v_n$  are column vectors of  $V$ .  $AV = VJ_n(\lambda)$  implies that

$$\forall i \in [n], (A - \lambda I)v_i = v_{i-1} \quad \text{where} \quad v_0 = 0.$$

Since  $(A - \lambda I)^{n-1} = V(J_n(\lambda) - \lambda I)^{n-1}V^{-1} \neq 0$ , it is trivial to find a vector  $x$  such that  $(A - \lambda I)^{n-1}x \neq 0$ . Set  $v_i = (A - \lambda I)^{n-i}x$ . It is sufficient to show the linear independence of  $\{v_i\}_{i \in [n]}$ . Suppose not, that is, there exists  $j \leq n$  such that

$$v_j = \sum_{i=1}^{j-1} a_i v_i \implies (A - \lambda I)^{n-j}x = \sum_{i=1}^{j-1} a_i (A - \lambda I)^{n-i}x \implies (A - \lambda I)^{n-1}x = \sum_{i=1}^{j-1} a_i (A - \lambda I)^{n-i+j-1}x.$$

Since  $(A - \lambda I)^n = V(J_n(\lambda) - \lambda I)^n V^{-1} = 0$ , we have that

$$(A - \lambda I)^{n-1}x = (A - \lambda I)^n \sum_{i=1}^{j-1} a_i (A - \lambda I)^{j-i-1}x = 0,$$

which contradicts our choice of  $x$ . Therefore,  $\{v_i\}_{i \in [n]}$  are linearly independent and  $V = [v_1, \dots, v_n]$ .

**(b)** Suppose there is only one eigenvalue  $\lambda$  and the Jordan blocks have sizes  $m_1 \geq \dots \geq m_k$ . We get the corresponding vectors for  $V$  as follows:

1. Initially  $i = 1$  and  $S = \emptyset$ .
2. Take  $x_i \in \ker(A - \lambda I)^{m_i} \setminus \ker(A - \lambda I)^{m_i-1}$  and  $(A - \lambda I)^{m_i-1}x_i \notin \text{span}(S)$ . Let the vectors corresponding to the  $i$ -th Jordan block to be

$$V_i = [(A - \lambda I)^{m_i-1}x_i, (A - \lambda I)^{m_i-2}x_i, \dots, (A - \lambda I)x_i, x_i].$$

3. Set  $S \leftarrow S \cup \{(A - \lambda I)^{m_i-1}x_i\}$ ,  $i \leftarrow i + 1$  and back to Step 2 until  $i > k$ .
4.  $V = [V_1, \dots, V_k]$ .

Note that such  $x$  always exists since  $\dim \ker(A - \lambda I)^{m_i} - \dim \ker(A - \lambda I)^{m_i-1} \geq i$  (there are at least  $i$  Jordan blocks whose sizes are at least  $m_i$ ) and  $\dim \text{span}(S) < i$  at step  $i$ . Finding  $x_i$  can be achieved by Gaussian elimination.

By **(a)**, the choice of  $V$  satisfies  $AV = V \text{diag}\{J_{m_i}(\lambda)\}_{i \in [k]}$  and for  $i \in [k]$ , the columns of  $V_i$  are linearly independent. Suppose the columns of  $V$  are not linearly independent, that is, WLOG, we have that there exists  $\mathbf{a} \neq \mathbf{0}$  (we can get rid of  $(A - \lambda I)^\ell x_i$  for  $\ell < m_i - 1$  by multiplying  $(A - \lambda I)^{m_i-1-\ell}$  on both sides),

$$\sum_{i=1}^k a_i (A - \lambda I)^{m_i-1}x_i = 0,$$

which contradicts our Step 2. In conclusion,  $V$  is a feasible choice. Also, note that for each column  $v$  of  $V$ ,  $v$  is the solution of  $(A - \lambda I)^{m_1 + \dots + m_k}u = 0$ .

For Jordan blocks with different eigenvalues, by Problem 3, we complete the proof by putting all  $V$ 's of the different eigenvalues together.

## Problem 5

We prove by induction on  $k$ . The statement follows from definition of the Jordan block if  $k = 1$ .

Now suppose  $k > 1$ . We have that

$$\begin{aligned}
(J^k)_{ij} &= \sum_{\ell=1}^n (J^{k-1})_{i\ell} (J)_{\ell j} = \sum_{\ell=i}^{\min(n, i+k-1)} \binom{k-1}{\ell-i} \lambda^{k-1-(\ell-i)} (J)_{\ell j} & (\text{I.H.}) \\
&= \lambda^{k+i-j} \sum_{\ell=i}^{\min(n, i+k-1)} \mathbb{1}[j-1 \leq \ell \leq j] \binom{k-1}{\ell-i} & (\text{def. of } J) \\
&= \lambda^{k+i-j} \times \begin{cases} \binom{k-1}{j-i-1} + \binom{k-1}{j-i} & , i \leq j-1 \wedge j \leq i+k-1 \\ \binom{k-1}{j-i} = 1 = \binom{k}{j-i} & , j = i \\ \binom{k-1}{j-i-1} = 1 = \binom{k}{j-i} & , j = i+k \\ 0 & , \text{o.w.} \end{cases}. & (2)
\end{aligned}$$

Since  $\binom{k-1}{j-i-1} + \binom{k-1}{j-i} = \binom{k}{j-i}$ , Equation (2) completes the proof.

## Problem 6

By definition we have that

$$\begin{aligned}
\frac{d(\det A(\mu))}{d\mu} &= \sum_{\sigma \in S} (-1)^\sigma \frac{d}{d\mu} \prod_{i=1}^n a_{\sigma_i, i} = \sum_{\sigma \in S} (-1)^\sigma \sum_{j=1}^n a'_{\sigma_j, j} \prod_{i \in [n], i \neq j}^n a_{\sigma_i, i} \\
&= \sum_{j=1}^n \sum_{\sigma \in S} (-1)^\sigma a'_{\sigma_j, j} \prod_{i \in [n], i \neq j}^n a_{\sigma_i, i} = \sum_{j=1}^n \det(a_1, \dots, a_{j-1}, a'_j, a_{j+1}, \dots, a_n).
\end{aligned}$$

## Problem 7

Block Diagonal Form (Corollary 1.5.45) shows the existence of  $T$ . We now prove the uniqueness property. Suppose  $T^{-1}AT = \text{diag}\{B_{ii}\}_{i \in [n]}$  and  $(T')^{-1}AT' = \text{diag}\{B'_{ii}\}_{i \in [p]}$  where  $B_{ii}$  has the same eigenvalues as  $B'_{ii}$  for  $i \in [p]$ . It is sufficient to show that  $T = T'V$  and  $V$  is an diagonal block matrix. Note that the invertibility of  $V$  follows from the invertibility of  $T$  and  $T'$ .

Let  $C = (T')^{-1}T$ . We write  $C = (C_{ij})_{i, j \in [p]}$  as a block matrix. We need to show that  $C_{ij} = 0$  for  $i \neq j$ . We have that

$$T \text{diag}\{B_{ii}\}_{i \in [p]} T^{-1} = (T') \text{diag}\{B'_{ii}\}_{i \in [p]} (T')^{-1} \implies C \cdot \text{diag}\{B_{ii}\}_{i \in [p]} = \text{diag}\{B'_{ii}\}_{i \in [p]} \cdot C. \quad (3)$$

Therefore, Equation (3) shows that

$$\begin{pmatrix} C_{11}B_{11} & \cdots & C_{1p}B_{pp} \\ \vdots & \ddots & \vdots \\ C_{p1}B_{11} & \cdots & C_{pp}B_{pp} \end{pmatrix} = \begin{pmatrix} B'_{11}C_{11} & \cdots & B'_{11}C_{1p} \\ \vdots & \ddots & \vdots \\ B'_{pp}C_{p1} & \cdots & B'_{pp}C_{pp} \end{pmatrix}.$$

Fix  $i, j \in [p]$  and  $i \neq j$ . Suppose  $V^{-1}B_{jj}V = R_{jj}$  and  $W^{-1}B'_{ii}W = R'_{ii}$  where  $R_{jj}$  and  $R'_{ii}$  are upper-triangular matrix and lower-triangle matrix respectively. Let  $H = W^{-1}C_{ij}V$ , we have that

$$C_{ij}B_{jj} = B'_{ii}C_{ij} \implies W^{-1}C_{ij}VV^{-1}B_{jj}V = W^{-1}B'_{ii}WW^{-1}C_{ij}V \implies HR_{jj} = R'_{ii}H.$$

Let  $\lambda_1 \cdots \lambda_{n_j}$  be diagonal entries of  $R_{jj}$  and  $\mu_1 \cdots \mu_{n_i}$  be diagonal entries of  $R'_{ii}$ . Since the eigenvalues of  $R_{jj}$  and  $R'_{ii}$  are disjoint,  $\{\lambda_k\}_{k \in [n_j]} \cap \{\mu_k\}_{k \in [n_i]} = \emptyset$ .  $HR_{jj} = R'_{ii}H$  implies the following equations

$$\lambda_1 h_{11} = \mu_1 h_{11} \implies h_{11} = 0, \quad h_{11}r_{12} + h_{12}\lambda_2 = \mu_1 h_{12} \implies h_{12} = 0 \cdots .$$

By the the argument in the class,  $H = 0$ . Since  $V, W$  are of full rank,  $H = 0$  implies  $C_{ij} = 0$ .